A review of recent advances and future trends in Wire Arc Additive Manufacturing

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Abstract. This article provides an overview of Direct Energy Deposition – Arc technology (DED-Arc), also known as Wire Arc Additive Manufacturing (WAAM), which involves the deposition of metal wire using an arc power source and a CNC or robotic manipulator. The high deposition rate of WAAM justifies its use for the manufacturing of small to large-size components with lower resolution and less complex geometry. However, the use of wire as feedstock in the WAAM process has certain advantages and disadvantages, which are explained in detail. The WAAM specialties are in-situ alloying and the production of functionally graded materials (FGMs). Various sensors, path planning, process control, and FEM simulation from WAAM are used to reduce material and energy consumption and make the process more sustainable. Post-processing techniques are also discussed as a method of improving the quality of the final product. Finally, the prospects of the WAAM process are presented.

Introduction

Wire arc additive manufacturing (WAAM) is an additive manufacturing method that uses standard welding equipment as a heat source and a robotic or CNC machine for movement. Improving the accuracy and repeatability of robots through the development of advanced controllers and CNC machines leads to the WAAM process with exceptional capabilities [1]. Currently, it is possible to produce large-scale components efficiently, cost-effectively, and environmentally friendly with WAAM, thanks to its high deposition rate, affordable equipment, and sustainable energy use [2].

In the aerospace industry, the ratio between the initial weight of the material block and the final weight of the finished product is known as the "buy-to-fly ratio" (BTF). Compared to the traditional manufacturing process, the WAAM process can reduce the BTF ratio from 10 to 20 to 1.2, which means that 220 kg of material can be saved in the production of a 24 kg component [3]. Owing to the complexity and welding nature of the WAAM process, a wide range of parameters can influence the final product, which necessitates the investigation of the manufacturing process, material, controll, modeling, and post-processing. Also, WAAM is not considered a net shape process, and machining is required to meet target dimension properties. This study provides information on the advantages and disadvantages of WAAM and the challenges and opportunities of its use for future research and development.

WAAM process

Wire arc additive manufacturing is categorized as Direct Energy Deposition (DED), and 3D components are fabricated layer by layer through melting consumable wire. At the end of 2018, the DED method took over 16% of the total AM market, and the WAAM process was even capable of

producing components with the certification of the US Federal Aviation Administration (FAA) [4]. In the WAAM procedure, the Arc Welding trajectory is followed by a robot, leading to an increasing dimension of the printed objects. The use of welding as the energy source makes this process more cost-effective than other processes, such as laser or electron beam AM. WAAM is an additive manufacturing method that uses standard welding equipment as the heat source and a robot or CNC machine for movement. Based on the feedstock type, DED is divided into powder-DED and wire-DED. WAAM is a welding-based deposition technology that is widely used in the industry because of its simplicity of use and high efficiency. The power efficiency of the welding process varies between 40% and 90%, depending on the process [5].

WAAM generally consists of a robotic moving unit, wire feeder, arc generator, gas shielding, and integrated control unit or PC. Gas metal arc welding (GMAW), gas tungsten arc welding (GTAW), and plasma arc welding are the three welding technologies that are often used in WAAM procedures [6]. GMAW, in contrast with GTAW and PAW, utilizes consumable electrode and wire that feed into the molten pool and solidify layer by layer. The polarity of the welding is an essential parameter in TIG welding; arc penetration, welding quality, and oxidation can be controlled by the polarity. Commonly for TIG welding, Direct Current (DC) with negative polarity is preferred for steel, while Alternating Current (AC) is typically employed for aluminum.

There are three types of metal transfer in the GMAW process: globular, spray, or short-circuiting methods, as shown in Figure 1. Spray transfer happens at a high current when small discontinuous droplets of metal move along the arc due to the electromagnetic force of the high frequency. At lower currents, the globular transfer takes place, whereby the size of the droplet increases to the diameter of the wire and falls into the welding zone due to gravity. Welding droplets in short circuit are small, and the solidification rate is high. However, the pinch effect makes weld poll neither stable nor spatter-free [7].

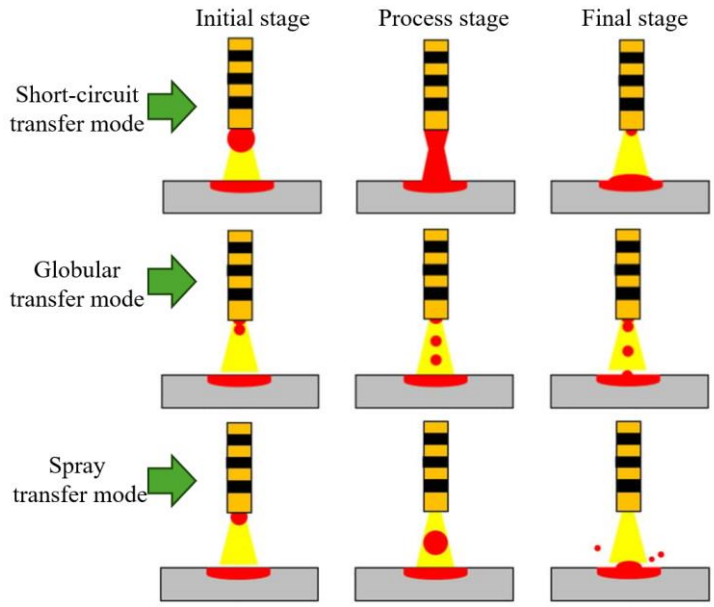


Fig. 1. Three different metal transfer types in welding droplets. Reprinted from [8] MDPI, under a Creative Commons Attribution (CC BY) License.

In 2004, the Austrian company Fronius developed Cold Metal Transfer (CMT), a process characterized by the continuous back-and-forth movement of the wire during the arc stage. This movement creates a short circuit, melting the filler wire into a droplet [9]. Also, to enhance the compatibility of the TIG with automation in robot usage to resemble the MIG process, Lincoln Electric's research center has developed a novel method called TOPTIG. This innovative approach aims to improve torch accessibility and facilitate the TIG process by feeding the wire to the highest temperature. Table 1 shows the properties of different WAAM processes.

Table 1. Different WAAM process and its properties [10].

WAAM process	Deposition rate	Samples size	Mechanical properties
CMT	High	Large	Low/medium (Anisotropic)
TOPTIG	Low	Small/medium	Medium/high (Homogenous)
PAW	Medium	Medium/Large	Medium/high

WAAM Parameters

To define appropriate parameters, process windows can correlate the travelling speed (TS) and wire-feed speed (WFS) in the WAAM process. The ratio of WFS/TS between 5 to 20 is suitable for the WAAM process [11]. As shown in Figure 2(a), Voltage, compared to the current, does not affect the welding penetration. Some of the WAAM variables are listed in Table 2.

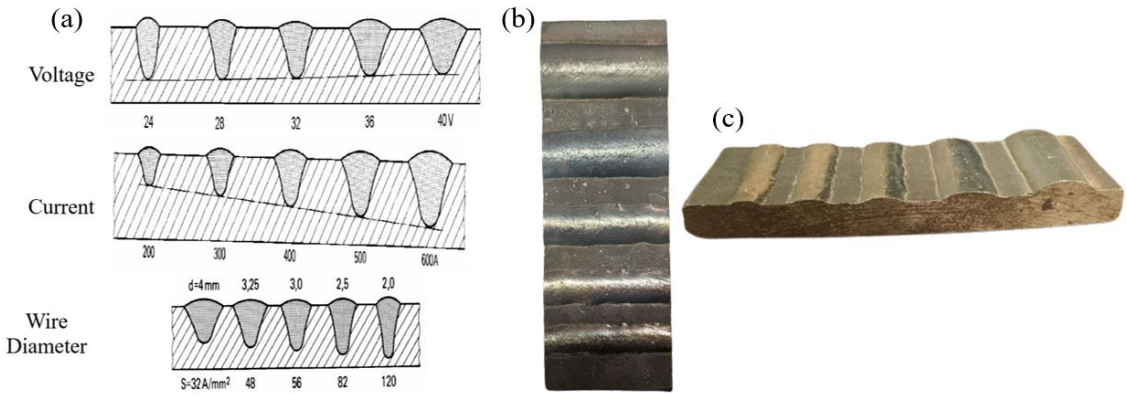


Fig.2. a) Voltage, current, and wire diameter impact on welding shape [12], b& c) welding shape change with heat input variation.

Table 2. Example of required parameters for the WAAM process.

Welding method	GMAW	TIG	PAW
Material	Al-1Mg [13]	Al-5Mg [14]	Carbon steel [15]
WFR (m/min)	Auto	1.6	3.15
TS (mm/s)	8	2.5 & 5	6
Arc current (A)	150	110	110
Voltage (V)	16.7	14 to 16	15.4
Heat input (J/mm)	Variable	180 to 210	-
Gas flow rate (L/min)	20	15	13.5 Argon, 0.5 Plasma
Shielding gas	Argon	Argon	Argon
Cooling time (min)	0 and 3	-	2
Arc length	-	5 mm	-
Tig Torch distance (mm)	-	-	6
Tig Nozzle dia. (mm)	-	-	3
Tig Tungsten setback (mm)	-	-	2.2

Freeform manufacturing in AM is of interest to the aerospace industry to exploit topology optimization and reduce the final product's weight. In addition to the importance of topological optimization of some components, there is an increasing need to significantly reduce material waste in the production process [3]. Using the WAAM process for laborious and complicated parts like

shipbuilding can open new eras to integrate Industry 4.0 to build components more effectively [16]. Vishnukumar et al. [17] repaired corroded AA5052 aluminum with ER4043 filler material for marine application. Lee et al. [18] utilized the WAAM process for repairing broken gray cast iron lathe machine components with nickel wire. Utilizing the WAAM process for structural and building applications is also approved with Eurocode 3 (EN 1993–1-1) standard for Union K 40 grade steel material [19]. As shown in Figure 3(c), the WAAM process can significantly reduce the repair time for propeller blades.

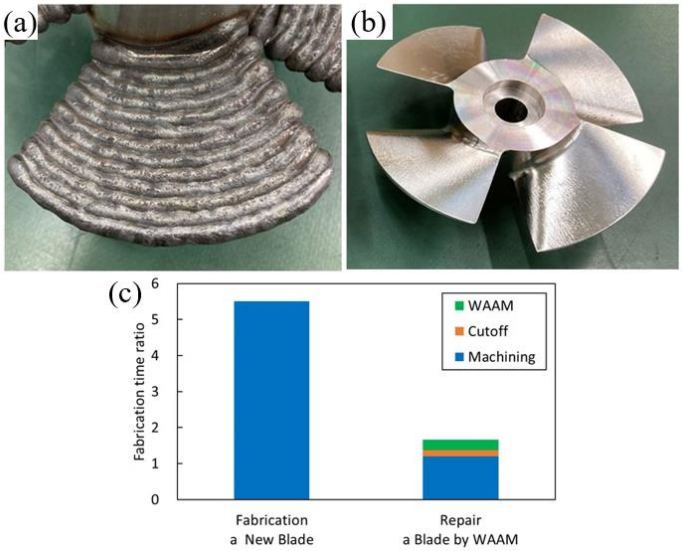


Fig.3. a) WAAM and near-net-shaped propeller blades, b) WAAM propeller blades after machining, c) Fabrication time for new blade vs repaired WAAM. Reprinted from [20] MDPI, under a Creative Commons Attribution (CC BY) License.

WAAM material

Single material WAAM

The material for the WAAM process is supplied in the form of wire spools, which are commercially available to produce a wide range of alloys. Materials such as titanium, steel, aluminum, and nickel find extensive application in the Wire Arc Additive Manufacturing (WAAM) process. Additionally, the development of multiwire WAAM systems allows combining these materials to produce specific and tailored material properties.

Titanium is not considered an easy material for casting due to its high melting temperature, reactivity with molding materials, component shrinkage, lower fluidity, and the need for special tools and costly methods to produce high-quality components. Due to the limitations above in the production of titanium, it is necessary to develop new methods like the WAAM process. The mechanical properties of the WAAM process are comparable to conventional and advanced manufacturing processes [21,22]. One of the main problems in the production of WAAM is the presence of oxidation due to the reactive properties of titanium at high temperatures. Oxidation-induced embrittlement can be mitigated by employing a closed chamber filled with inert gas, although this restricts the movement of the robot, as shown in Figure 4 [23]. Magnesium alloys have great potential for use in mobility applications to achieve weight reduction due to their high strength. Consideration of the WAAM process and its properties with respect to topology optimization has revealed the further advantages of using magnesium [24].

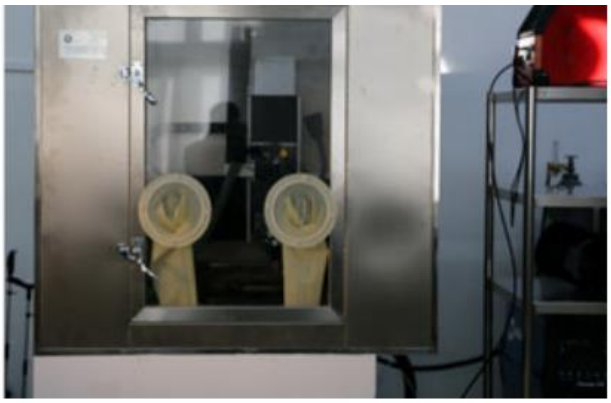
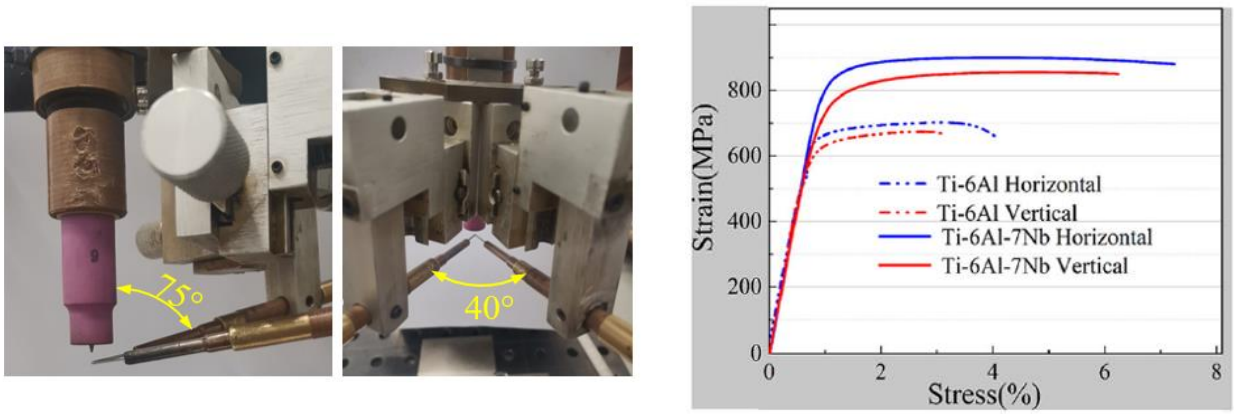


Fig. 4. Developed chamber for WAAM process of titanium. Reprinted from [25], MDPI, under a Creative Commons Attribution (CC BY) License.

Multi-material: In situ alloying

Production efficiency in the WAAM process can be improved using the double WAAM process instead of the single wire process. The concept of in-situ alloying involves careful selection and precise control of the added alloying elements. Yang et al. [26] utilized twin wire arc additive manufacturing (T-WAAM) with Nb foil to fabricate Ti-6Al-7Nb alloy. The deposited alloy displayed a distinctive Widmannstatten structure and periodic heat-affected zones (HAZ). Figure 5 illustrates the in-situ alloying of titanium alloys and their mechanical properties.



(a) (b)

Fig.5. a) In-situ alloying of Ti-6Al-7Nb in WAAM b) stress-strain curves of in-situ alloyed. Reprinted from [26], Taylor & Francis, under a Creative Commons Attribution (CC BY) License.

To calculate the mass fraction (E), the below equation can be used [27]:

$$E = \frac{\sum WFS_i D_i^2 \rho_i E_x}{\sum WFS_i D_i^2 \rho_i} \quad (1)$$

Where Ex represents the proportion of a specific element (Cu, Mg) in each wire, WFSi refers to the wire feed speed, and Di represents the diameter of the wire. ρi represents the density of the wire. Shen et al. [28] successfully fabricated 30 at% Al iron aluminides in the GTAW process with pure Fe and Al wire. The microstructures show the epitaxial development of larger columnar grains.

Multi-material: FGM

Functional graded materials (FGM) are an advanced class of materials having unique qualities that vary spatially in a regulated way. Wire arc additive manufacturing (WAAM) is a cost-effective and precise technique highly suitable for producing functional graded materials (FGM). Using WAAM to deposit multiple materials, FGMs can be created with controlled material properties. Furthermore,

WAAM is a potential approach for the fabrication of FGM because of its benefits, such as high deposition rates and accuracy. A major difficulty in multi-material processing is the formation of cracks in the area where the different materials meet, known as the material transition zone (MTZ). These cracks are typically caused by brittle phases that result from the mixing of other materials or from differences in their coefficients of thermal expansion [29].

Defects in the WAAM Process

While there are drawbacks associated with fabricating components using WAAM, such as suboptimal surface quality, limited dimensional accuracy, and occasional low density, it is essential to consider additional factors, such as slow cooling rates and the significant heat source induced by the process [30,31].

Due to the high heat input with WAAM, distortion of the components can happen due to residual stresses. The following steps could be done to reduce the amount of residual stresses in WAAM [3]. Procedures like symmetrical structure, back-to-back structure, optimization of component orientation, and inter-pass rolling with pressure can reduce residual stresses. The clamped sample produces the highest thermal stresses, and the annealing effect of the previously deposited layers is evident as the amount of stress is reduced [32]. Cold-work processes have various advantages, including a decrease in porosity, an increase in material quality, a reduction in stress and distortion, an increase in part accuracy, and an improvement in mechanical qualities [33]. Rolling in the WAAM process of AZ31 can decrease residual stresses up to 14.32% [34]. Figure 6 shows various defects in the WAAM production of the AZ31 alloy.

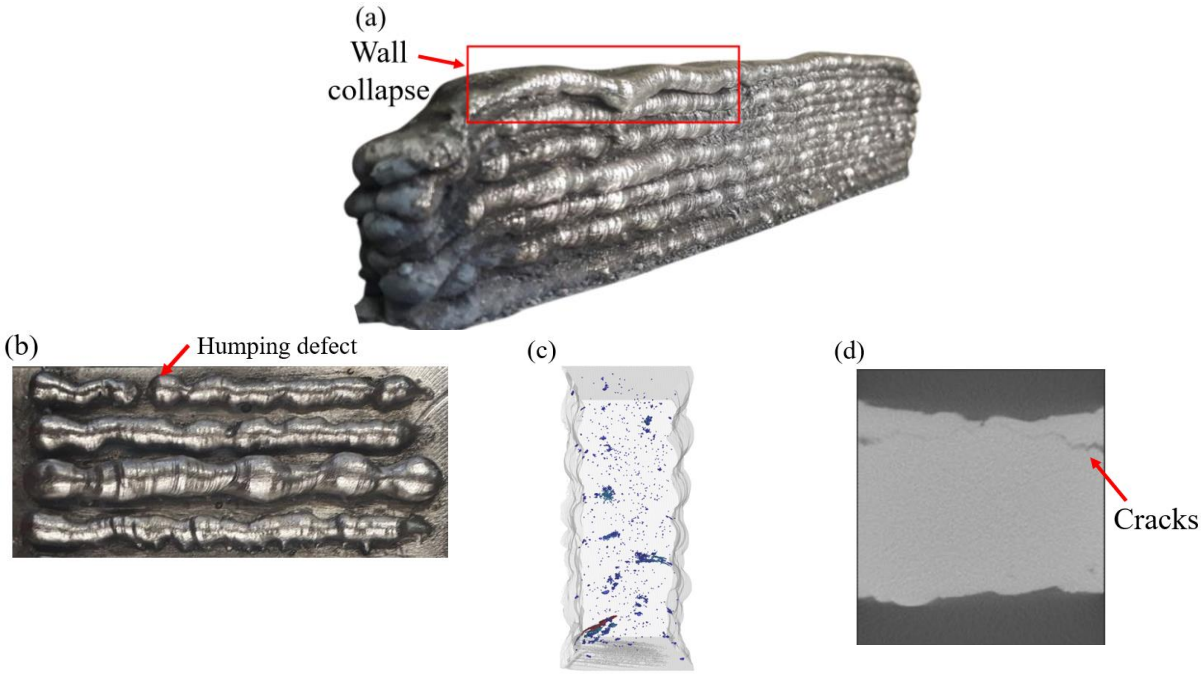


Fig.6. Defects in the WAAM process of AZ31, a) Wall collapse, b) humping defect, c) XCT scan of porosity formation, d) cracks.

Magnesium is highly reactive with oxygen and nitrogen at high temperatures, which can lead to pore defects in fused magnesium alloys, especially in open atmospheres containing water and oxygen. The rapid cooling of magnesium, as well as its low density and low buoyancy force, make it difficult for bubbles to rise and escape during solidification in magnesium and aluminum components [24]. In addition to porosity and residual stresses, various issues can arise during the WAAM process, such as humping [35], solidification and Liquation cracking [36], delamination [37], balling [38], inconsistent and variable thickness [39], the star-step effect [40], oxidation [41], spatter [42], and

non-uniform microstructure [43]. The printing window is shown in Figure 7 for WAAM AZ91D with possible defects [44].

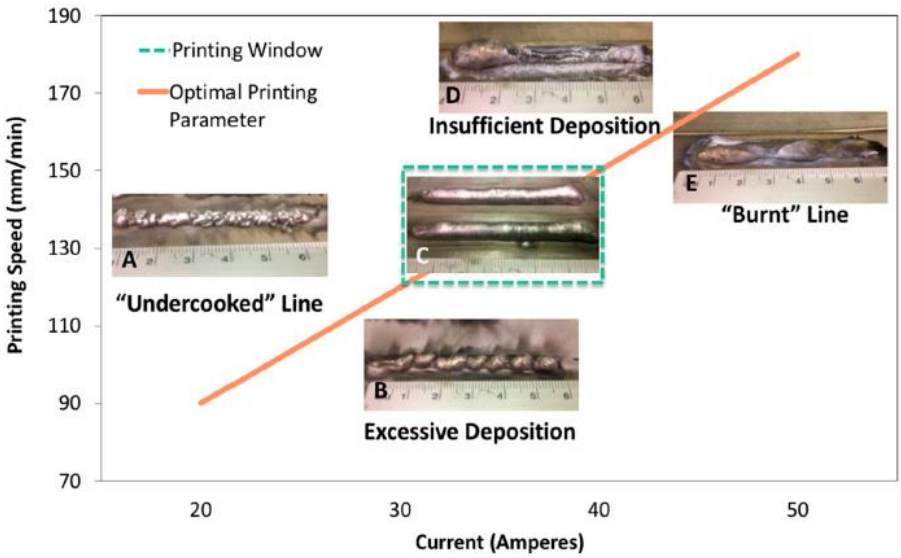
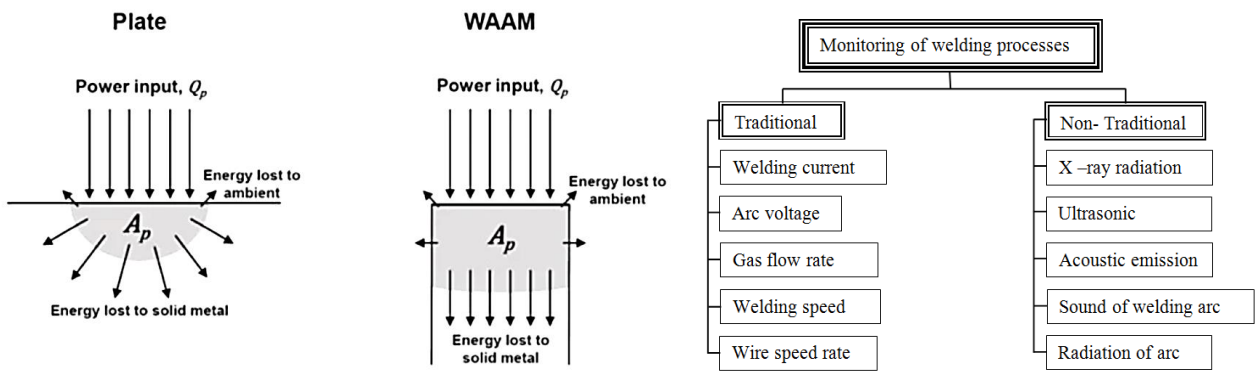


Fig.7. a) Printing window of AZ91D and probable defects. Reprinted from [44], MDPI, under a Creative Commons Attribution (CC BY) License.

WAAM Process planning

WAAM is a promising arc welding-based method for the aerospace industry, offering the potential to reduce buy-to-fly ratios from 20:1 with forging and machining to 5:1 with WAAM [45]. However, the process of WAAM, in contrast to other additive manufacturing methods like powder bed fusion, needs to be developed more to produce components with reliability, dimensional precision, and homogenous material properties. The complexity of the process and insufficient process control are the main obstacles to providing consistent deposition as a result of the complex thermal cycling involved in arc-based metallurgy in WAAM [46].

Due to the high heat input in WAAM, the components are more susceptible to distortion, residual stresses, and geometrical inaccuracy, which necessitate monitoring development. As shown in Figure 8(a), heat conduction tends to propagate in several directions under general boundary conditions, whereas heat conduction in the WAAM process is in one direction [47]. Since WAAM is similar to welding, the monitoring process of WAAM, according to Węglowski [48], can be divided into two sections, as shown in Figure 8(b). Traditional monitoring is mainly related to the system rather than produced components like voltage and current.



(a) (b)

Fig.8. a) Thermal conductivity in WAAM and plate. Reprinted from [47], MDPI, under a Creative Commons Attribution (CC BY) License. b) Classification of monitoring for welding

adopted from the welding process. Reprinted from [48], IntechOpen, under a Creative Commons Attribution (CC BY) License.

Traditional monitoring

The heat input is the main parameter of the welding operation and has an essential impact on the distortion and geometrical accuracy as follows:

$$Heat\ input = \frac{Voltage \times Current \times 60}{Welding\ speed}\ (J/mm) \tag{2}$$

The current has the most critical impact on the fusion penetration and weld geometry, while voltage is related to the reinforcement and fusion zone dimension [6]. The layer height of the WAAM is not constant and varies from one layer to another. The arc current plays a vital role in determining the contact tip-to-work distance (CTWD). Ščetinec et al. [49] controlled the WAAM layer height through current monitoring. Using short arc welding can reduce the input energy of the welding process. At constant voltage, an increase in CTWD leads to an increase in resistance and a decrease in welding current, as depicted in Figure 9 [50].

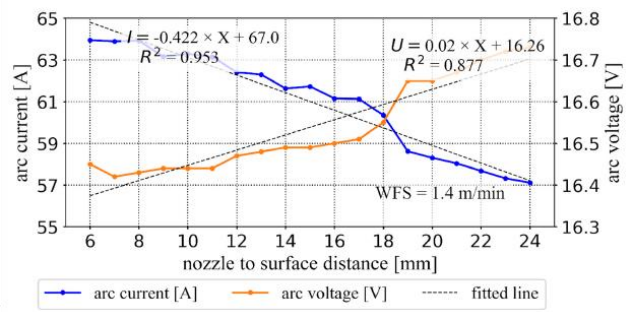


Fig.9. Current and voltage variation with CTWD, Reprinted with permission from Elsevier from [49].

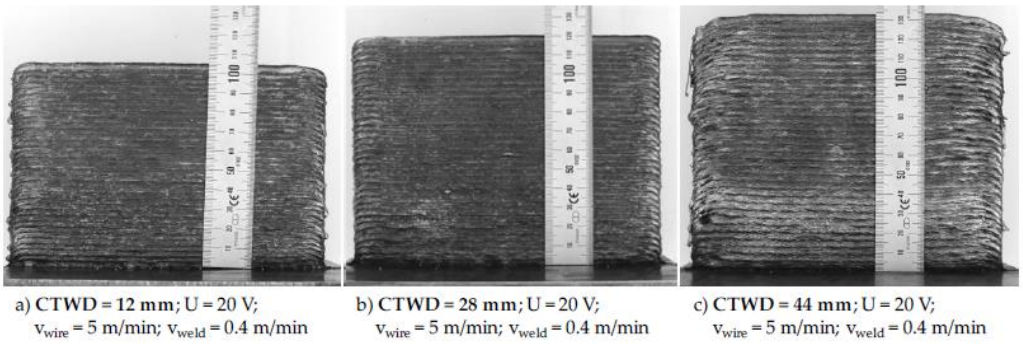


Fig.10. Height change with different CTWD. Reprinted from [50], MDPI, under a Creative Commons Attribution (CC BY) License.

Non-Traditional monitoring

The generation of sound signals accompanies the welding process of WAAM and contains information about the arc specification, the molten pool, and the droplet transfer [52]. The irregularity of the weld bead is associated with the current and acoustic signals, and the process instability from high-frequency arc oscillations results in sound signals [53]. Chabot et al. [54] controlled CTWD with sound monitoring. Ramalho et al. [55] introduced deliberate contamination of the WAAM wall and exploited short-time Fourier transform and Power Spectral Density to study the acoustic signals.

Computer vision in the welding industry is mainly utilized for data acquisition, dimension measurement, defect detection, object tracking, and calibration. Compared to welding, WAAM requires several layers to be applied, which results in a more complicated process. Deposition of bead

and width is an efficient way to use the material in WAAM. Xiong et al. [56] applied an online controller based on segmented neuron self-learning to maintain bead width for thin-walled structure stability. Xiong et al. [57] proposed a new controlling method of deposition height by measuring the current and previous layer, applying error to the PID controller, and changing the wire feed speed. Wang et al. [58] used a 3D scanner with a camera to control the width and reinforcement in WAAM via current variation. Kozamernik et al. [59] integrated fast cooling with an IR pyrometer to measure the inter-pass temperate and successfully printed a 3D shape with a 2% deviation.

In-process and post-process operations in WAAM

Traditional subtractive manufacturing is less efficient than the WAAM approach, and using the post process in WAAM is more favourable regarding material and energy consumption. According to the component's size, WAAM can save fabrication time by 40–60% and post–machining time by 15–20% [2]. Figure 11 demonstrates three main steps of the WAAM process.

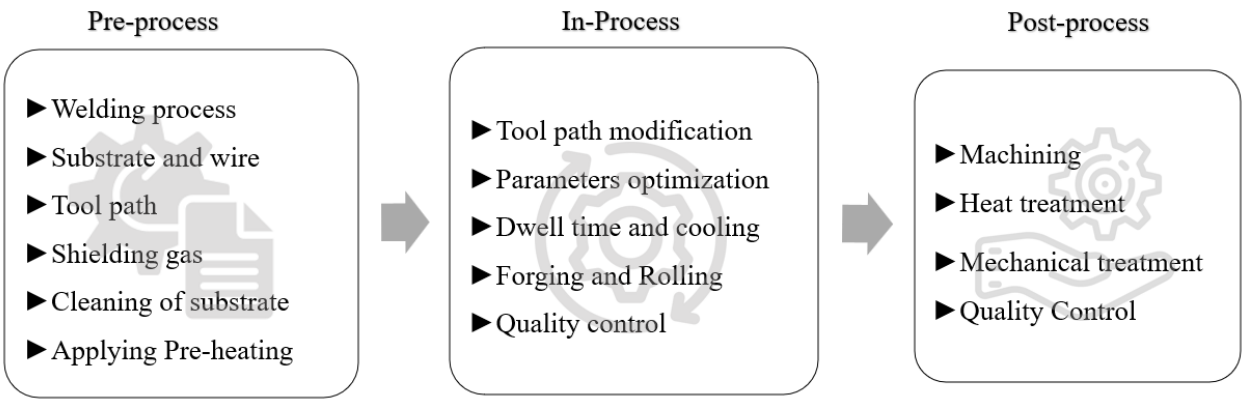
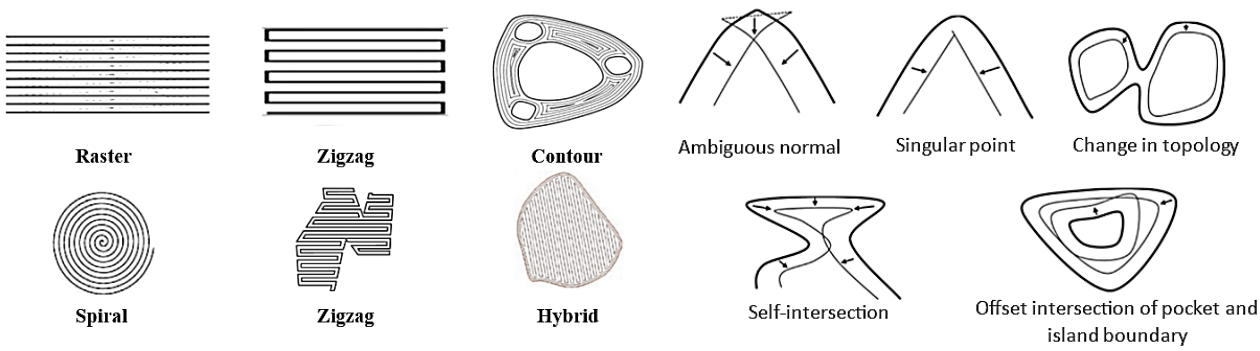


Fig.11. Process flow in WAAM

Pre-process: Tool path generation

Due to the limitations imposed by the weld bead and layer geometry, path planning in WAAM is more complex than in powder-based Additive Manufacturing. A wide range of path planning has been developed, as illustrated in Figure 12. Development in processing techniques like using GPU instead of serial method can speed up the process of zigzag generation problems up to 9 times faster [60]. Singh et al. [61] employ a Travelling Salesman Problem (TSP)-based algorithm to optimize toolpath generation, minimizing retractions and turns. Giordano et al. [62] introduce a novel continuous manufacturing path, eliminating the stop and start of welding.

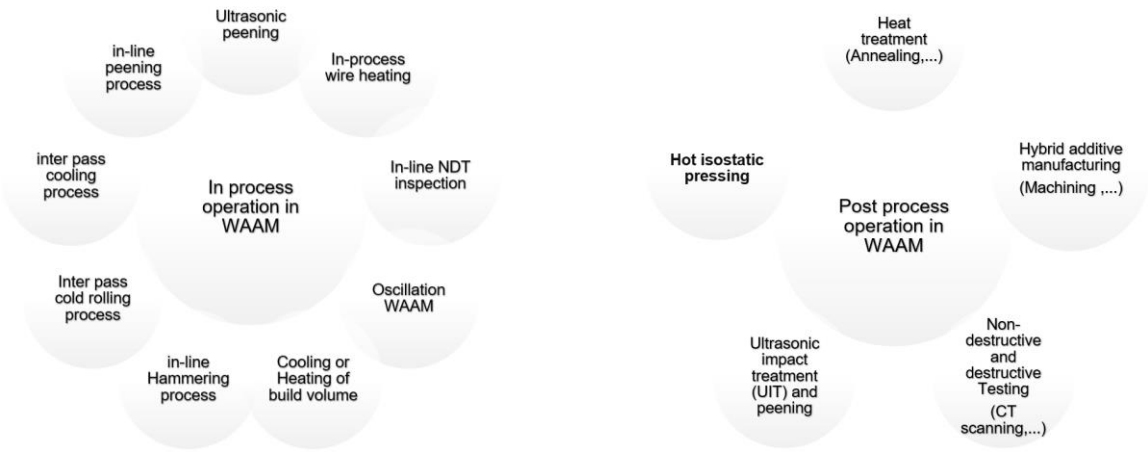


(a) (b)

Fig.12. a) Common path planning in the WAAM process b) Common challenges in contour offsetting, Reprinted from [63], MDPI, under a Creative Commons Attribution (CC BY) License.

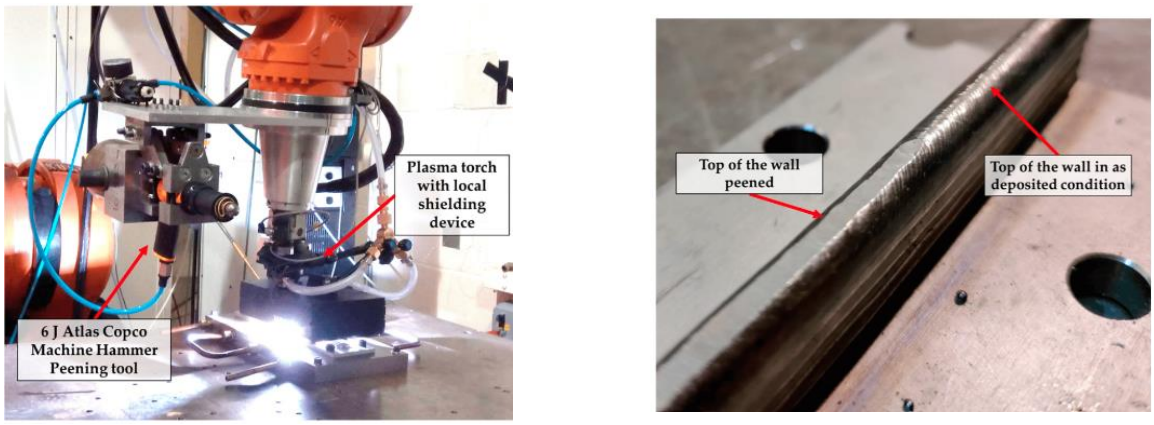
In-process operation in WAAM

In WAAM, cold rolling is a process to relieve residual stresses and deformation by reducing anisotropy and improving the microstructure. This procedure enhances mechanical characteristics. The characteristic microstructure of components produced through WAAM generally exhibits large columnar grains, resulting from repeated cycles of melting and solidification during the process. To obtain grain refinement, hybrid WAAM processes such as inter-pass rolling (IRWAAM), inter-pass machine hammer peening (IMHP, figure 14), ultrasonic impact treatment (UIT), ultrasonic peening treatment (UPT), laser shot peening (LSP), active inter-pass cooling device (AIC), magnetic arc oscillation (MAO), hotwire (HWAAM) has been developed. Figure 13 demonstrates common in-process and post-process operations for the WAAM. Increasing the radius of the roller protrusion increased the size of the recrystallized region, and the more significant rolling pressures generated more refinement [64]. Duarte et al. [65] modified a novel hot forging technique to WAAM of Stainless Steel AISI 316L known as HF-WAAM. In this method, the regional Forging process occurs rapidly after deposition at high temperature, contributing to viscoplastic deformation. The cooling rate plays a key role in the microstructure and mechanical properties of WAAM products, and the addition of argon, CO2, nitrogen, or forced water cooling can improve grain refinement [66].



(a) (b)

Fig.13. a) In-process, and b) post-process operation in WAAM.



(a) (b)

Fig.14. a) Machine hammer peening (MHP) is in the WAAM process, and b) as deposited vs. peened wall. Reprinted from [67], MDPI, under a Creative Commons Attribution (CC BY) License.

Post-process operation in WAAM

Post-processing in the WAAM operation is generally divided into heat and mechanical treatment, machining, and quality control. WAAM is a process that aims to achieve near-net shape production, but it typically requires a final machining step. The development in robotic machining can lead to integrated machining in the WAAM process, and compared to wrought titanium, the WAAM specimens were found to be more machinable [68]. Not only is post-heat treatment time-consuming and costly, but the size of the furnace can also limit the size of the components in the WAAM process. In the WAAM process, the repetition of thermal cycles implies that the subsequent cycles can be considered as post-treatment for the previously applied layer. It has been proven that new layers can have a stress-relief effect on the previously applied layers [69]. Reference [70] applied the T6 heat treatment process in the WAAM process of aluminum and declared that a high concentration of small hydrogen pores formed due to the precipitation of hydrogen micropores.

In no way could traditional, tactile, or optical measurement systems analyse the non-accessible surfaces of the additive manufacturing parts due to the complexity of the geometry. Computed Tomography (CT), as a non-destructive method, has an undeniable advantage regarding the inspection time of the AM components since it can conduct the metrology and material evaluation in one process and is time-saving. CT scans have enormous applications in additive manufacturing, such as specification of defects, dimensional metrology, and texture inspection of internal surfaces [71].

WAAM simulation

The interlayer cooling time in the WAAM process is related to the time-temperature-transformation (TTT) diagram to prevent the formation of unwanted phases. The influence of component shape, deposition method, and substrate's dimensions contribute to the difficulty in forecasting the cooling time. Viola et al. [72] investigated the manufacturing time to merge the thermal and kinematic criteria for the WAAM process. Ogino et al. [73] numerically developed a model for the WAAM process of GMAW and studied the effects of the welding conditions on the deposited layers investigated. They showed that inter-pass cooling influences the formation of higher and thinner deposition. The variation in deposit height was reduced by reversing the welding direction in each layer. Manurung et al. [74] used a coupled thermomechanical solution with phase-change occurrences determined by latent heat in the numerical simulation. They simulated heat source movement with Goldak’s double ellipsoid model through hexagonal elements for meshing. Figure 15 depicts the simulation of the WAAM process and rolling process.

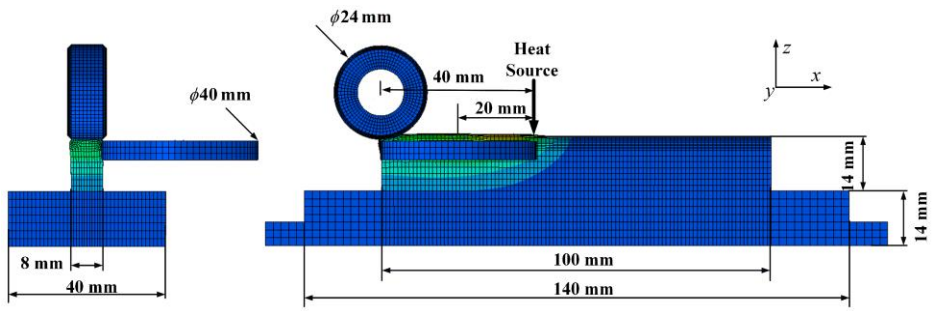


Fig.15. FEM model of rolling for WAAM [75], MDPI, under a Creative Commons Attribution (CC BY) License.

Conclusion

A comprehensive overview of the current technical advances of the WAAM process was given, focussing on microstructure, mechanical properties, defects, in-process and post-process treatment. WAAM is a solid option that can be used instead of conventional manufacturing, especially to produce medium to large sized components. WAAM can be performed with different arc welding

technologies, resulting in different material properties. The parameters, material properties, and process planning have a major influence on the properties and shape of the final WAAM component.

The WAAM process enables complex process enhancements such as FGM and in-situ alloying to be carried out, offering greater flexibility and the potential for the development of advanced materials at lower cost. In-process and post-process operations like hammering and forging can improve the mechanical and microstructural behaviour of WAAM products.

Due to the instability of the thermal process in WAAM, it is necessary to develop advanced monitoring systems and utilise simulations that can save time, material, and energy, ultimately leading to more sustainable additive manufacturing. The WAAM process is prone to defects that require further research and development, especially in non-destructive testing.

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